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Points to Consider No. 14

Manufacturing of ATMPs – Facility Design (Part 1)

2025



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Drug
Association®

Connecting
People, Science
and Regulation®

pda.org

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Points to Consider No. 14

Manufacturing of ATMPs – Facility Design (Part 1)

2025



ISBN: 978-1-945584-51-0

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Table of Contents

Introduction	6
Abbreviations	9
Topic 1. Risk Management	10
Q 1.1. How can a risk assessment be created for an ATMP facility?	10
Q 1.2. How does the ATMP risk assessment determine the biosafety level requirement?	13
Topic 2. Facilities	15
Q 2.1. How do you design facility process flow to minimize extraneous contamination and cross-contamination and prevent errors in an ATMP facility?	15
Q 2.2. What should be considered for open and closed systems to achieve the required product and operator protection for ATMPs?	18
Q 2.3. What is common practice related to the pressure concepts for maintaining classification and containment for ATMPs?	19
Q 2.4. What should be considered when designing cleanrooms, HVAC, air locks, and corridors for handling products such as viral vectors in an ATMP facility?	22
Q 2.5. What environmental risk factors are to be considered outside of the manufacturing facility (e.g., neighborhood of the manufacturing facility, hospital, clinic, home setting) when working with highly potent or genetically modified organisms?	24
Topic 3. Utilities	25
Q 3.1. Are there any specific requirements for designing utilities in an ATMP facility?	25
Topic 4. Process Equipment	29
Q 4.1. When choosing a biological safety cabinet for the manufacture of ATMPs, what factors are important?	29
Q 4.2. What factors are important when choosing an isolator for the manufacture of ATMPs?	33
Q 4.3. What should be considered when comparing biosafety cabinet and isolator design?	37

Topic 5. Supplementary Technical Information	42
Q 5.1. What is important to consider in a mock-up for biosafety cabinets and isolators?.....	42
Q 5.2. Are there any differences in the frequency of chamber leak-integrity testing procedures between an isolator and a biosafety cabinet?	46
Q 5.3. What are the similarities and differences between a biosafety cabinet and an isolator in the objective, procedure, documentation, and expectations of an airflow visualization (smoke study)?	47
Q 5.4. What methods are used to maintain facility cascade/pressure stability?.....	51
Recommended Reading	53
References	56

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Figures and Tables Index

Figure 2.3-1 Open Isolator for Aseptic Filling with Continuous Transfer of Primary Packaging (e.g., vials in and out).....	20
Figure 2.3-2 Example of a Closed Isolator Used for Cell and Gene Manufacturing (i.e., with no small openings).....	20
Figure 2.3-3 Sample Air-flow Diagrams: (left) Biosafety Cabinet with Grade B Surrounding Area; (right) Closed Isolator with Grade D Surrounding Area.....	21
Figure 4.1-1 Biosafety Cabinet	30
Figure 4.1-2 Airflow Pattern in a Biosafety Cabinet	31
Figure 4.2-1 Closed Isolator with Material Air Lock, Centrifuge, and Incubator	34
Figure 4.2-2 An Example of a Closed Isolator System for the Manufacturing of ATMPs	34
Figure 4.2-3 Computational Fluid Dynamics Studies	35
Figure 4.2-4 Safety Filter Before the Air-return Ducts.....	35
Table 4.3-1 Background Grades and Assurances Considerations	37
Table 4.3-2 Comparison Between Biosafety Cabinet and Isolator for Facility Design	39
Table 4.3-3 Capital Expenditure and Operating Cost Considerations.....	40
Table 4.3-4 Cost Comparison Between the Use of Biosafety Cabinet and Isolator.....	40
Figure 4.3-5 Isolator and Biosafety Cabinet Environment Options (positive-pressure cascade and bubble in the air locks).....	41
Figure 5.1-1 Computerized Mock-Up	43
Figure 5.1-2 Virtual-Reality Mock-Up.....	44
Figure 5.1-3 Physical Mock-Up	44
Figure 5.3-1 Containment Test Method on a Biosafety Cabinet (KI-Discus-Test)	49
Figure 5.3-2 Containment Test Method on a Fume Cupboard (ConT [®] Test)	50
Figure 5.4-1 Shows the Pressure Difference Measurement Between the Isolator Chamber and Above the Supply HEPA Filter.....	51

Introduction

Advanced Therapy Medicinal Products (ATMPs) are unique because they are based on cells, genes, or tissues from animals and human beings. ATMPs are produced either in low-batch volumes, such as products produced for an individual patient (autologous), or (relatively) larger batch volumes for products produced for multiple patients (allogeneic).

ATMPs are classified into three main types.

1. **Gene Therapy Medicines:** These contain genes that lead to a therapeutic, prophylactic, or diagnostic effect. They work by inserting “recombinant” genes into the body, usually to treat a variety of diseases, including genetic disorders, cancer, or other long-term diseases. A recombinant gene is a segment of DNA that is created in the laboratory, bringing together DNA from different sources.
2. **Somatic-Cell Therapy Medicines:** These contain cells or tissues that have been manipulated to change their biological characteristics, or cells or tissues not intended to be used for the same essential functions in the body. They can be used to cure, diagnose, or prevent diseases.
3. **Tissue-Engineered Medicines:** These contain cells or tissues that have been modified so they can be used to repair, regenerate, or replace human tissue.

“Combined ATMPs” are ATMPs that combine with one or more medical device(s) as an integral part of the medicine.

In general, ATMPs should adhere to the same manufacturing requirements as sterile parenteral drugs for their upstream, downstream, and fill-and-finish requirements. The minimum expectations for the manufacture of ATMPs are essentially the same as for all biological process considerations, with some specific nuances depending on the starting material, end product, intended administration window, and intended recipient(s). The use of ATMPs for clinical trials must also be manufactured under equivalent quality standards to the manufacturing of ATMPs with a marketing authorization **(1-3)**.

The basic good manufacturing practice (GMP) principles of all processes include design, control, and monitoring. For ATMPs, these are:

- **Design**
 - An appropriate Pharmaceutical Quality System (PQS)
 - A detailed design of the process (open, closed, or hybrid, for autologous and/or allogeneic therapies)
 - Facilities, utilities, and process equipment that support the process design
 - Prevention measures to avoid extraneous contamination, cross-contamination, error, or mix-ups

- Material, personnel, and waste flows that support the process and prevention measures
- **Control and Mitigation**
 - A formal contamination control strategy (CCS) specific to the process
 - Control of starting materials and their source/donation and critical raw materials that are appropriately sourced and controlled for quality
 - Manufacturing environment controlled in a manner that supports the CCS
 - Laboratory practices and personnel training that support the PQS
 - Process (Safety) equipment controlled in a manner that supports both the PQS and CCS
 - Effective Pharmaceutical Quality System (including a facility risk assessment)
 - Traceability requirements that support batch identity and traceability from manufacture through distribution
 - Warehousing, distribution, and temperature controls that support product quality throughout all phases of production and distribution
- **Monitoring**
 - Process-environment-related attributes in accordance with CCS
 - Cleanliness of facility and equipment to support fulfillment of CCS
 - Testing (including microbial, particulate, airflow, and HVAC) to confirm integrity of environmental controls
 - Release-testing strategy to ensure fulfillment of the quality target product profile (QTPP)

This document considers the facility, utilities, and process equipment designs necessary to support the manufacture and testing of ATMPs (autologous and allogeneic). Ideally, the design considerations should be weighed upfront to ensure the process meets the respective requirements.

ATMP processes are still an emerging science. As a consequence, there are a variety of groups that are involved in their development and commercialization including academia, research organizations, hospitals, and biopharmaceutical companies. At the time of compilation of this Points to Consider (PtC) document, various manufacturing environment technologies are employed in the manufacture of ATMPs. Products are commonly manipulated in biosafety cabinets (BSCs), isolators, or closed systems with the required cleanroom surroundings. The stage of commercialization does not always dictate the type of facility or type of technology through which the products are produced; rather, the environment or technology is often selected based on the flexibility required for the process.

These technologies all have controls to ensure the environment where product manipulations occur is maintained in a manner that prevents contamination. These cell-, gene-, or tissue-engineering-based products are classified from occupational safety requirements to have a working environment determined by its Biological Safety Level (BSL).

ATMP manufacturing includes some products that are meant to replace, repair, or modify genes in a cell. To achieve this, certain viral vectors (e.g., adeno-associated virus, retrovirus, lentivirus, herpesvirus) are used for that purpose. Viral vector-based products can vary in their BSL depending on the type of virus and its potential to cause disease in humans. These are commonly classified as BSL-1 or BSL-2 and, in rare circumstances, even higher (e.g., BSL-3), depending on the virus load and type of virus. This Points to Consider document will focus on BSL-1 and BSL-2 requirements only, as BSL-3 and BSL-4 facilities are rare for the manufacture of ATMPs.

Product and process characterization is performed using quality risk management (QRM) principles to help the sponsor understand the importance of each aspect of the developed process/product. This information includes requirements for environmental conditions, production equipment/systems, product quality, and occupational safety. These elements have been reflected in GMP guidelines, norms, and practices for decades. Additional recommendations on these concepts specific to ATMPs are addressed in this technical document.

Given the novelty of these products and the relative scarcity of comprehensive technical guidance in support of manufacturing, it can become difficult to find resources to help determine appropriate strategies to ensure the successful manufacture of ATMPs. This Points to Consider (PtC) is an initiative of the PDA ATMP Advisory Board and was compiled by a diverse team of PDA subject matter experts to address some of the common questions within the industry regarding the manufacture of these products.

The manufacture of ATMPs should generally comply with the GMP and regulatory guidance for other biologic and aseptic products. Therefore, many of the topics that are included in this document may overlap with the conventional handling of biologics. However, the scarcity of comprehensive technical guidance and common questions from the industry seeking help to understand how various requirements apply to ATMP processes necessitate this guidance exhaustively addressing those design, control, and monitoring requirements, even when those requirements overlap with conventional processing.

As technology advances and regulatory expectations for continuous process improvement continue to grow, pressures may increase within the industry to further minimize the risk of our processes by moving to greater containment, greater sterility assurance, and reduction of personnel involved in the process. This is not always possible with the current state of the art for these novel processes. Hence, this PtC addresses the common technologies employed today and presents the risk-benefit analysis for each so that manufacturers can realistically assess their current process while simultaneously understanding alternate designs and controls that may be available for implementation.

This document is organized into topics and various subtopics that address the questions raised on the subject of ATMP manufacture. Each question first presents an overall recommendation, which may be followed by a rationale for the recommendation and/or appropriate references for further reading.

Abbreviations

ATMP	Advanced Therapy Medicinal Products	HVAC	Heating, Ventilation, and Air Conditioning
BSC	Biosafety Cabinet	MAL	Material Air Lock
BSL	Biological Safety Level	Non-RCV	Non-Replication-Competent Virus
CCS	Contamination Control Strategy	PAL	Personnel
CFD	Computational Fluid Dynamics	RCV	Replication-Competent Virus
EM	Environmental Monitoring	RTP	Rapid Transfer Port
GMO	Genetically Modified Organisms	VPHP	Vapor Phase Hydrogen Peroxide
GMP	Good Manufacturing Practice	WFI	Water for Injection
HEPA	High Efficiency Particulate Air		

Topic 1. Risk Management

Q 1.1. How can a risk assessment be created for an ATMP facility?

Recommendation

The manufacture of any product, including an ATMP, necessarily entails some degree of risk. The degree of risk often correlates with such factors as the number of manual interventions, the type of environment in which those interventions occur, and the open or closed nature of the process. For example, autologous therapies produced in small batches (small-scale) often have a higher risk per unit produced when manual manipulations are numerous. The facility design element of a formal facility risk assessment should always be included as part of the Design Review process in the organization, but it is even more important when processes have varying levels of risk. Based on the process design principles and nature of the work to be performed in the facility, the organization can define the risk profile of the facility and work to mitigate those risks through appropriate design controls associated with the equipment (e.g., isolators, BSCs) and appropriate process improvements (e.g., single-use systems, closed processing, automation, improved intervention design). As part of the same facility risk assessment, design controls can also be considered to manage risks related to biological hazards.

Facility risk assessment should be undertaken by an interdisciplinary team and should be guided by a risk-assessment facilitator of appropriate experience along with a scribe to record the risk assessment. The scope of the facility risk assessment should include both facility and equipment design and selection, as well as product, personnel, and environmental risk categories. Including personnel and environmental safety aspects in the risk assessment is recommended as it is likely that safety-related controls could also have an impact on product quality. Control measures for one risk category should be assessed for impact on other risk categories. For example, the use of a virus in cell therapy manufacturing may pose health risks to personnel while facilities producing the viral vectors have additional safety considerations due to the active replication of viral vectors, albeit in a controlled process and environment. Where applicable, specialized safety personnel, such as those with biosafety, toxicology, microbiology, and virology expertise, should be involved in the performance of the risk assessment and in the design of the mitigations of the identified risks.

Facility risk assessments should consider, but are not limited to, the following elements:

1. Material

- a. Presence of adventitious agents (e.g., potential presence of replication competent agents).
- b. Material handling and transfer into and out of critical process areas (e.g., appropriate design and classification of pass-through containers, surface decontamination/disinfection during transfer of frozen materials, transfer of sterile parts/consumables)

- c. Controls and material-flow requirements when viruses, cellular materials, or genetically modified organisms (GMOs) are present in the facility (including raw material, product, sample, and waste flows)
 - d. Incoming control of materials for quality, microorganisms, and adventitious agents (e.g., vendor qualification, vendor audit, vendor certification, in-house sampling and testing, controls at loading dock, warehouse)
- 2. Facility and equipment**
- a. Equipment, material, and waste flow (e.g., material or equipment moving to a higher-grade area, cross-contamination due to non-unidirectional flow)
 - b. Personnel flow (e.g., cross-contamination risk due to personnel working or supporting different batches or different products at the same time)
 - c. Cleanability of equipment and facility, including contact materials (e.g., selection of materials of construction that are nonadditive, non-absorptive, or non-reactive, including primary equipment, gaskets, filters, and selection of product-/process-compatible lubricants)
 - d. Need for dedicated, segregated, self-contained areas and separation by time (e.g., manufacturing process using infectious viral vectors)
 - e. Containment adequacy (e.g., ability to contain potential aerosol release, examination of open processing for possible routes of contamination)
 - f. Radiation shielding (e.g., use of x-ray irradiation equipment in process may require additional consideration directed by local regulations)
 - g. Aspects of commissioning, qualification, maintenance, and environmental monitoring (EM) (e.g., handling of pass-through containers, process gases)
- 3. Multiproduct processing** (e.g., handling multiple batches of product and/or multiple products simultaneously to ensure chain of identity to avoid mix-ups, limit risks of cross-contamination, prevent processing errors, and ensure process adherence)
- a. Pressure-cascade design, which meets the containment design and aseptic processing requirements for all materials, especially highly potent materials and the appropriate controls for transitioning from area to area (e.g., air lock and pass-throughs with appropriate interlocks and purge times before opening to other zones or classifications)
 - b. Utilities that ensure appropriate quality, reduce risk of contamination, ensure adequacy of supply and appropriate control (e.g., steam, process gases, power, clean compressed air, instrument air, heating, ventilation and air conditioning (HVAC), water for injection (WFI) and/or purified water, liquid nitrogen, carbon dioxide)
- 4. Cleaning, decontamination, disinfection, sanitization, and sterilization**
- a. Equipment and facility cleaning/decontamination design (e.g., design of automated decontamination, cell-virus inactivation process)
 - b. Waste decontamination (e.g., solid or liquid, in-process or waste treatment system)
 - c. Accident and spill response, decontamination, and cleaning
 - d. Equipment and material disinfection, sanitization, and sterilization
- 5. Airflow design**
- a. Airflow within the facility design should consider the acceptability of recirculation of air within the facility or whether single-pass air (i.e., no recirculation) is required
 - b. Zone control within the HVAC air-handling units should consider separation by cleanliness and/or by process
 - c. Differential pressure management to ensure appropriate containment and/or cleanliness

6. External environment

- a. Impact of activities performed in the facility on the external environment (e.g., waste treatment requirement, construction)
- b. Seasonal changes of incoming air and source water, where applicable, to ensure that adequate pretreatment is included in the utility system design
- c. Pest control measures

Rationale

A facility risk assessment can help determine the most appropriate design of facilities and selection of equipment. In particular, it is important to consider not only quality risks but also environmental and safety risks. Examples of how hazards can affect both safety and quality aspects include, but are not limited to:

1. Certification and qualification test for a BSC covers product, personnel, and environmental risk but placement and layout of BSC should also be verified according to manufacturer and/or standard requirements (e.g., NSF/ANSI 49). The layout will then likely impact personnel, material, sample, and waste flows and potentially affect the cross-contamination control strategy. If the risk cannot be adequately controlled with the use of a BSC, a closed system or isolators should be used.
2. Waste treatment is part of the biological risk management in the facility, and the type of waste treatment will impact personnel and material flow and potentially affect the cross-contamination control strategy.
3. Facility decontamination processes may leave residues that subsequently impact product quality and patient safety but may also represent a risk to operator safety.
4. Cross-contamination between manufacturing areas of different products (e.g., live vs. inactivated microorganisms) will impact product quality and patient safety, as well as put operators at risk.
5. Design requirements for BSL containment of GMOs or other highly potent products may conflict with aseptic processing requirements. For example, a negative-pressure cascade may be recommended to contain GMO within the facility, but aseptic processes rely on a positive-pressure cascade to prevent contamination from entering the facility.

Q 1.2. How does the ATMP risk assessment determine the biosafety level requirement?

Recommendation

A list of biological materials, including GMOs, should be maintained and reviewed regularly against local requirements as part of a risk management program. The list serves as a basis for the organization to determine the risk group classification of each material, and the risk group is determined by the hazard posed by the biological agents and toxins (if any) used in the manufacturing process. Although risk group classification depends on localities, the definition is generally in line with those from the World Health Organization (WHO) and U.S. Food and Drug Administration (FDA) (4, 5):

- **Risk Group 1** (no or low individual and community risk)
 - A microorganism that is unlikely to cause human or animal disease.
- **Risk Group 2** (moderate individual risk, low community risk)
 - A pathogen that can cause human or animal disease but is unlikely to be a serious hazard to laboratory personnel, the community, livestock, or the environment. Laboratory exposures may cause serious infection, but effective treatment and preventive measures are available, and the risk of the spread of infection is limited.
- **Risk Group 3** (high individual risk, low community risk)
 - A pathogen that usually causes serious human or animal diseases but does not ordinarily spread from one infected individual to another. Effective treatment and preventive measures are available (not considered in this document).
- **Risk Group 4** (high individual and community risk)
 - A pathogen that usually causes serious human or animal disease and that can be readily transmitted from one individual to another, directly or indirectly. Effective treatment and preventive measures are not usually available (not considered in this document).

Upon understanding the risk group classification of the materials involved in the product, the organization should consult with the local environmental health and safety regulatory bodies and the inspecting regulatory bodies of the countries in which the product is licensed to determine the BSL requirements for the manufacturing facility. The organization can then determine the controls required to manage the risk using facility risk assessment. Typically, risk group classification and control requirements are heightened for the following situations:

- Novel biological materials (e.g., new gene insertion or deletion, novel virus)
- Biological material that is replication competent as opposed to biological material known to be no longer capable of replication
- Viral vectors and their risk to the operator and cross-contamination

The organization should also consider the requirements related to the use of GMOs. Many countries maintain specific facility and procedural requirements for facility-handling of GMOs, and the containment requirements are typically applied in addition to the typical pharmaceutical facility cleanliness requirements.

Risk management principles should be applied when determining the control points. The risk assessment and mitigations should aim to minimize the risk to as low as is reasonably practical.

Current work related to the cell and gene therapy industry generally uses materials that are classified as Risk Group 1 or Risk Group 2 and are rarely in the higher classification (e.g., 3 or 4).

Discussing the BSL requirements with the applicable regulatory bodies as early as possible is recommended to ensure regulatory concerns are addressed in the facility design stage.

Rationale

Process and product knowledge are important factors to enable the application of science and a risk-based approach to determine what controls to apply to the manufacturing facility.

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Topic 2. Facilities

Q 2.1. How do you design facility process flow to minimize extraneous contamination and cross-contamination and prevent errors in an ATMP facility?

Recommendation

Facility design plays an essential role in minimizing contamination, cross-contamination, and prevention of errors. For this purpose, the possible risk factors that can lead to contamination and cross-contamination should be evaluated and defined in the CCS document.

Contamination and cross-contamination occur through several different vectors, which include but are not limited to:

Physical

1. Contaminated packaging or containers – For example, a packaging format of any raw or starting materials used in the process that cannot be adequately decontaminated upon transfer for use.
2. Mechanical transfers – For example, spillages of materials, through improperly sealed transfer devices, or through risks such as the “ring of concern,” which is an area on a rapid transfer port that cannot be adequately disinfected and may be exposed during transfer procedures if not addressed.
3. Personnel – For example, contamination of clothing from non-process-dedicated clothing regimes.
4. Waste processes.
5. Airborne sources – For example, Aerosols, failure of air-handling units in the Grade A environment, or volatile constituents of contaminated waste.

In order to minimize these risks, a containment strategy should be developed from the inside-out (from critical operations to less-critical operations) based on a risk assessment. The containment strategy and CCS should identify any critical manufacturing processes, paying particular attention to higher-risk operations.

Based on the risk related to the primary protection and containment measures, a secondary means of protection and containment should be planned, such as the personnel air lock (PAL) and material air lock (MAL) pressure cascades (e.g., from production room to air lock and from air lock to the corridor) (see **Figure 4.3-1**). Personnel, product, sample, and waste paths should also be considered to avoid possible contamination in other areas and to minimize the risk of cross-contamination.

Important Considerations

1. Is the overall process closed, such as a closed process in a BSC or an open process in a closed isolator? Does it remain closed throughout the whole procedure? If not, can a fully closed process be integrated?

- a. Any process which can be performed closed should be designed and executed closed to minimize the risks associated with open processes.
 - b. Items required within the Grade A space (e.g., tools, equipment) should be sterilized. When sterilization is not possible, the requirement for the continued use of that item should be evaluated and, when deemed necessary, the highest possible level of surface disinfection should be pursued with appropriate controls and monitoring and justified within the CCS to ensure risk to the process is minimized.
 - c. Intrinsic sterile connectors should be used, when possible, to avoid multiple aseptic connections.
2. How are transfers in and out of the BSC or isolator performed? Do they have closed transfers via rapid transfer ports (RTP) or within a MAL where an automated surface decontamination is executed?
3. Isolators or BSCs should use appropriate technologies like RTPs for transferring sterilized or disinfected material to reduce the risk of physical transfer of contaminants.
4. How is the pharmaceutical product handled inside the BSC or isolator? Does the handling pose a risk of contamination and, if so, how can this risk be eliminated?
 - a. First-air protection of any open-handled processes inside the BSC or isolator.
 - b. Greatest possible separation of personnel from the process through the use of barriers and sterile implements.
5. Can the packaging material for the pharmaceutical product be externally contaminated before leaving the isolator? If so, what engineering controls could be used to eliminate this contamination?
 - a. The use of air locks compatible with vapor phase hydrogen peroxide (VPHP) or other appropriate decontamination/inactivation should be used, if possible. Otherwise, an RTP or other closed containment system should be used to transfer the material to a safe area where the decontamination/ inactivation takes place.
6. What are the methods to manage pharmaceutical-product residues or waste (e.g., liquid waste, solid waste)?
 - a. For processes requiring removal of waste during the operation, the waste should be handled in a contained manner, through a separate pathway from product or product samples, so that it does not become a source of contamination.
 - b. If the product exit is not separate from the waste exit, the removal-of-waste operation should be performed in a contained manner at a dedicated time separate from the product manufacture.
7. How will the processing area be cleaned to eliminate the risk of cross-contamination?
 - a. Any inactivation step of residual pharmaceutical product should be performed before cleaning.
8. Are single use closed systems used? How are cross-contamination risks considered?
 - a. Contamination can occur at the interfaces of closed single-use systems, such as single-use bioreactors or single-use bags, both during the process (e.g., during additions or sampling) and when they are dismantled if the systems and their interfaces are not designed appropriately.
 - b. The tightness/integrity should be tested and validated for the single-use system. This includes both vendor certifications and potentially in-house testing when that testing can be performed in a manner that is safe for the process, operator, and environment.

9. How will process leaks and spills be handled and addressed?
- Consider the containment measures to be taken to stop and prevent the spread of the leak/spill contamination. Ensure that there is adequate protection for air returns that are near the affected leak/spill area.
 - Establish the necessary and appropriate inactivation/decontamination steps which do not put product or packaging at risk and create tools that help operators determine what steps should be performed as soon as the leak/spill occurs and what can wait until after processing has concluded. (See **Item 6.**)
 - Consider the necessary handling of any process equipment, containers, or packaging that may have become contaminated on the outside. (See **Item 4.**)
 - Establish waste-handling procedures for any materials that were used to contain, inactivate, or clean up the leak/spill. (See **Item 5.**)
 - Establish procedures for personnel involved in the remediation of the leak/spill to ensure gloves or gowning that may have been impacted by the leak/spill or the clean-up are prevented from any additional spread of contamination (e.g., the requirement to change affected gowning).
 - Establish deviation and investigation procedures to assess the product/process during which the leak/spill occurred.

Additionally, to reduce the risk of intrinsic particulate contamination and maintain low mold in support areas, such as supply and return corridors, kitting of consumables should be performed in, at minimum, an ISO-8 cleanroom or controlled non-classified area to prevent introduction of particulate matter from the warehouse. Warehouses should be maintained free of cardboard and paperboard by transferring materials into totes prior to storage. This lower-particulate warehouse for materials destined for Grade B cell-processing suites should be separated from the main warehouse where non-product-contact items are stored separately by rollup doors, ideally with air-wash curtains.

All of these points and the risks they involve should be considered in a CCS document in relation to cross-contamination.

Rationale

An appropriate closed process, with a cleaning and inactivation strategy, minimizes the chance of extraneous contamination and cross-contamination of the product and contamination of surrounding areas and personnel. Furthermore, the risk of possible cross-contamination from the tertiary to the remainder of the facility will also be minimized. Well-documented procedures and training will also contribute to the prevention of cross-contamination.

Q 2.2. What should be considered for open and closed systems to achieve the required product and operator protection for ATMPs?

Recommendation

The use of closed systems offers distinct advantages. They are often used to scale out processes, once the process has been developed, and can often be performed appropriately within either a BSC or an isolator (although an isolator offers greater protection in the event of improper handling, spills, or leaks). The integrity of the closed system ensures a sterile environment. Hybrid or open processes within a closed isolator offer protection of both the product and the operator, providing the process steps and manipulations are performed correctly. Open processes within an open BSC require further implementation of mitigation measures to ensure no extraneous contamination or cross-contamination occurs.

Open System

An open system is one where the material is exposed to the manufacturing environment for even a short time. Due to this exposure, the environment is required to be of Grade A cleanliness. The manufacturing environment is a total sum of the events that occur during the process and any particulate or bioburden, pyrogenic, or viral contamination present in the area has the potential to contaminate the product. Therefore, each and every action using an open system has the potential to be the vector that carries the contamination. This is why the environmental controls and monitoring, movement of the operators, effectiveness of material transfers, and sanitization steps are of key importance.

Open systems are often used for early-phase/proof-of-concept trials when the required manipulations are still being defined and refined.

Closed System

A closed system is hermetically sealed from the manufacturing environment inside which the product pathway is considered sterile. Examples include single-use systems that have been gamma-irradiated that use intrinsic sterile connectors.

Systems are only considered closed when they have been integrity-tested and have been demonstrated to be hermetically sealed. Fully closed systems can be deployed and used in a Grade C environment if adequate control measures are implemented to avoid the risk of contamination.

Hybrid System

Hybrid processes, such as those requiring an open addition into a closed system, or that are open during initial assembly but are closed thereafter, are required to be performed within a Grade A cleanroom with Grade B background if using a BSC, or within Grade D background when using a closed isolator (depending on the CCS requirements). The increased cleanliness requirement for the surrounding environment is due to the fact that hybrid processes may create the opportunity for exposure of the product pathway to the environment.

Q 2.3. What is common practice related to the pressure concepts for maintaining classification and containment for ATMPs?

Recommendation

Regulations and Interpretations

The leading regulatory standards for aseptic manufacturing include:

- FDA *Guidance for Industry: Sterile Drug Products Produced by Aseptic Processing—Current Good Manufacturing Practice*
- European Union (EU) *Annex 1: Manufacture of Sterile Medicinal Products, EudraLex – Volume 4 – EU Guidelines for Good Manufacturing Practice for Medicinal Products for Human and Veterinary Use*

These guidances clearly state that pressure differentials for supporting clean areas must be maintained in order to keep sterile processing areas clean: that is, free from any contaminant (viable or nonviable).

Background Environment

Aseptic core operations in isolators or in BSCs are conducted in Grade A areas and, therefore, require pressure cascades to the background environment. In the case of a BSC, the surrounding external environment is a minimum of Grade B. For an isolator operation, the external environment is a minimum of Grade C for open isolators when they are connected with small openings for a semi-continuous or continuous material transfer in/out (**Figure 2.3-1**), or Grade D for a closed isolator (**Figure 2.3-2**, **Figure 4.2-1**, and **Figure 4.2-3**).

Cleanrooms are designed to maintain positive pressure, preventing "unclean" (contaminated) air from flowing inside. The idea is to ensure that filtered air always flows from the cleanest to less-clean spaces (i.e., a pressure cascade). EU GMP Annex 1 states that a positive pressure differential of a minimum of 10 Pascals should be maintained between each adjacent space (**I**). In a multichambered cleanroom, for example, the cleanest room is kept at the highest pressure. Pressure levels are set so that the cleanest air flows into spaces with less-clean air (e.g., air locks that act as a bubble or as a sink). Thus, multiple pressure levels may need to be maintained. For potent products or for viral vectors, the pressure concept should be adapted to prevent spreading and cross-contamination into surrounding areas by using negative and positive adjacent areas. .

The BSC is an open system, therefore the pressure differential cannot be regulated in the same way as an isolator. Instead, the surrounding environment is controlled with respect to the air locks and corridors but also to prevent contamination in the BSC Grade A area. The airflow between the corridor, air locks, and manufacturing area could be designed according to the examples shown in **Figure 2.3-3**. **Figure 4.3-1** contrasts a BSC cleanroom to a closed-isolator cleanroom, which includes containment as part of its design.

MALs, in particular, and PALs should be designed with interlocks that prohibit opening both doors at once. They should also have an integrated timer that enforces the required time in seconds to ensure that at least four to five (4-5) room air-changes occur within the MAL to reduce particles introduced into the higher-classified side as well as to control contamination.

Cleanroom suites should be designed for vapor-phase hydrogen peroxide (VPHP) or chlorine dioxide (ClO₂) compatibility. For product changeover (different vectors) and for replication-competent virus (RCV) to non-RCV as well as improving EM performance, whole-suite fumigation should be carried out periodically. RCV is a crucial safety concern, especially in gene therapy, which often uses modified viruses as vectors to

deliver therapeutic genes into cells. These vectors are engineered to be replication-defective (unable to reproduce on their own), but there is a risk that the virus could revert to a RCV during the manufacturing process due to recombination events. If this happens, the RCV could replicate uncontrollably in the patient's body, potentially causing serious health problems.



Figure 2.3-1 Open Isolator for Aseptic Filling with Continuous Transfer of Primary Packaging (e.g., vials in and out)



Figure 2.3-2 Example of a Closed Isolator Used for Cell and Gene Manufacturing (i.e., with no small openings)

Small Scale, ATMP Manufacturing Air-flow Diagram

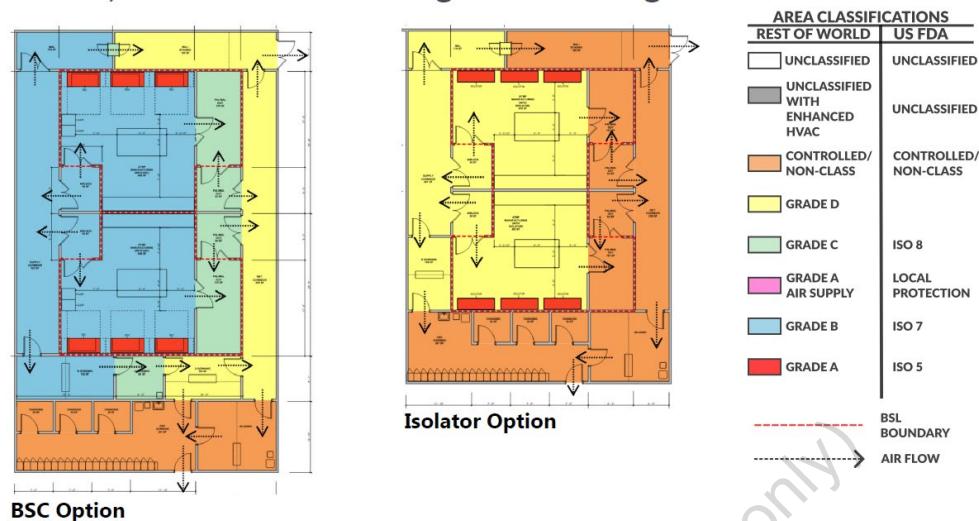


Figure 2.3-3 Sample Air-flow Diagrams: (left) Biosafety Cabinet with Grade B Surrounding Area; (right) Closed Isolator with Grade D Surrounding Area

Note: Design depends on the product and local regulations.

Rationale

Maintaining differential pressures (i.e., pressure cascades) is important to providing the highest protection to the product. The highest pressure should always be present in the most critical area and then flow to the less-critical areas, for example, from the isolator or BSC to the room. Air locks can be designed to have a higher pressure than both the room and the corridor to avoid contamination in the corridor, when concerned about containment of the process.

Q 2.4. What should be considered when designing cleanrooms, HVAC, air locks, and corridors for handling products such as viral vectors in an ATMP facility?

Recommendation

When designing a production site for ATMPs, a variety of factors should be considered, including, but not limited to the required process, protection of the product, prevention of cross-contamination, and protection of operators. The potency of the manufactured product and the inherent risk associated with manufacturing components (e.g., viral vectors) should be considered. Good engineering design practices also consider such lifecycle factors as initial installation, operation, maintenance/calibration, and eventual replacement and retirement strategies. In addition to including them in the engineering design features within the facility, these factors should also be part of a comprehensive CCS for operations, monitoring, and maintenance of the facility. The majority of these elements are not only requirements for ATMP production, but they are also applicable to traditional biotechnology or traditional cleanroom operations. Some essential elements include:

Design

- Fulfilling a minimum of a Grade D background (closed isolator) or Grade B background (BSC) with appropriate graded air locks for material and personnel.
- Designing all cleanrooms to be easily cleaned and disinfected; however, as some of the ATMPs require BSL-1 or higher for manufacturing, the rooms design should take into consideration the specific procedures for cleaning and/or inactivation required by the specific BSL.
- Designing all equipment and room features to be hygienic and engineered to withstand cleaning and disinfecting/inactivation agents used in the facility. For this reason, cleaning, sanitizing, and inactivation procedures and chemical compatibility should be considered early in the design of the manufacturing facility.
- Installing appropriate methods (e.g., autoclaves), based on the required BSL, to deactivate/decontaminate waste or leftover raw materials/products before leaving the area adjacent to the filling cleanroom.
- Consider locating emergency response equipment (e.g., emergency shower, eye shower) outside the cleanroom, according to the required potency or BSL, while ensuring there are no open drains located within the classified environments.

HVAC

- Designing the HVAC system to be appropriate for the product potency and/or the required BSL.
- Using high-efficiency particulate air (HEPA) filtration to obtain the required cleanroom classification.
- Designing the air supply for each space to promote appropriate air pressure differentials between rooms of varying classification or based on required operations (e.g., for maintenance of cleanliness or containment).
- Designing the installation and replacement of HEPA filters so they can be easily maintained in a way that minimizes cross-contamination and permits periodic integrity testing.

Air Locks

- Positioning air locks to separate spaces of different classifications and/or to avoid spreading contaminants in adjacent areas such as corridors. Designing PALs and changing rooms so that the exiting operators do not contaminate the entering operators (i.e., providing separate routes of entry and exit).

- Separating MALs from PALs.
- Designing MALs to allow for decontamination procedures within their environment to permit the movement of materials from a lower grade to the higher grade, or between a process space and a general use space of the same classification.
- Restricting personnel from crossing through a MAL to enter the processing cleanrooms or corridors. Instead, consider designing a procedure for personnel to deliver material to one side of the air lock and then, after decontamination, to retrieve that material from the other side of the air lock.
- Allowing waste materials to pass through MALs/PALs when clean materials or personnel are not present.
- Separating the waste flow from personnel and material in/out, based on the potency and BSL requirements of the products.

Corridors

- Plotting personnel, material, samples, and waste flows throughout the facility to ensure that corridors and common spaces are designed to promote unidirectional flow where possible (e.g., cleanest to “dirtiest”) and to avoid risks associated with intersecting flow paths that might permit cross-contamination of products or contamination of product or personnel with waste. (See **Figure 2.3-3**.)
- Designing the facility to employ clean corridors with positive pressure from the corridor to the MAL and PAL or the MAL and PAL with positive pressure to the corridor (e.g., air locks as bubbles) if the product is highly potent or if multiproduct manufacture will occur in the same facility, (e.g., air locks as sinks) based on the CCS. (See **Figure 2.3-3** and **Figure 4.3-1**.)

Rationale

The design considerations of the facility for the manufacture of ATMPs require careful in-depth understanding of the intended processes and the personnel, materials, waste, and sample flows that pertain to those processes. The adjacent rooms, corridors, air locks, and HVAC all play a part in ensuring the product is free from contamination or cross-contamination from other processes and provide for the safety of the operators. Additional safety requirements might be needed based on the determinations from a risk assessment and should be captured within the CCS.

Q 2.5. What environmental risk factors are to be considered outside of the manufacturing facility (e.g., neighborhood of the manufacturing facility, hospital, clinic, home setting) when working with highly potent or genetically modified organisms?

Recommendation

Highly potent materials and GMOs are considered major environmental risk factors in facility design since there are often additional local regulatory requirements to prevent unwanted release of such materials to the environment. The organization should consult local regulatory requirements, including assessing the biological materials handled in the facility against such regulations, including GMOs where appropriate.

In addition, the organization should work with the facility that is the point of administration to the patient (e.g., hospital) to determine any containment control requirements (e.g., administration precautions, patient waste, hospital waste). A discussion with the local regulatory authorities is recommended since ATMP technology is developing rapidly, and novel technology may result in novel products with novel risks.

Research laboratories are familiar with the use of novel technology, and biosafety risk is typically managed through the oversight of an Institutional Biosafety Committee (IBC). The use of ATMPs with biosafety concerns in healthcare settings has resulted in the extension of the IBC's role into patient administration settings, which is not within the traditional purview of conventional IBCs.

With regards to the risk associated with the use of highly potent or gene therapy products in a healthcare facility, a contact precaution over a 30-day period post-product administration, including waste handling, is recommended. Similarly, when the patient has returned home, their families are recommended to take similar precautions for the identical duration (6). Disinfection and cleaning procedures, including the relevant disinfectant contact time, must be shown to be appropriate for the organisms and/or materials involved in the ATMP within the post-administration setting (e.g., the home).

One particular risk associated with the use of ATMPs in healthcare settings is related to the ability to ensure an appropriate level of handling competency. Healthcare settings tend to include the use of floating staff where the personnel assigned are not always involved in one particular activity. Furthermore, the training of healthcare personnel (e.g., physician, nurse) is managed differently than for production workers; for example, the use of a learning management system may not be common in a healthcare setting and, therefore, tracking specific training competency may be difficult.

Rationale

Safety concerns for highly potent materials or GMOs must be addressed regardless of the setting in order to protect patients, healthcare workers, home caregivers, and others who may come into contact with the patient. The evolving nature of ATMPs means that the systems designed to control these extended risks are not fully mature. Therefore, it is incumbent upon the manufacturer to provide appropriate guidance for the necessary controls.

Topic 3. Utilities

Q 3.1. Are there any specific requirements for designing utilities in an ATMP facility?

Recommendation

There are no specific design requirements for utilities used for the manufacture of ATMPs compared to traditional small-molecule or large-molecule manufacture. However, specific requirements pertaining to containment and BSL facilities may apply according to local and international regulations. EU GMP Annex 1 and other suggested readings provide recommendations of utilities for product contact. Although the design requirements are generally consistent with other types of manufacture, the following good principles apply to clean utilities:

1. Gaseous disinfection system

- a. A risk-based commissioning approach for gaseous disinfection systems should be applied.
- b. Airtightness of the space needs to be demonstrated if gaseous disinfection is designed into an ATMP facility to decontaminate the facility/rooms. All surfaces and/or penetrations (e.g., sensors, electrical sockets) and openings that cannot be sealed in routine operation (e.g., the gap around the door, leak-tight isolation dampers in the HVAC duct) should be adequately sealed to prevent the escape of the decontaminating agent(s).
- c. The performance of the gaseous-disinfection system should be demonstrated through validation and appropriate Environmental Monitoring Performance Qualification.
- d. The residual level of disinfectant post-disinfection process should be validated using trace-limit-detection methods to demonstrate that the residuals are removed to levels that will have no adverse impact on the manufactured pharmaceutical product.
- e. Pipes should be marked with gas identity signs and should indicate the flow direction.

2. Carbon dioxide generation and distribution system

- a. Any carbon dioxide (CO₂) distribution lines installed should be tested to confirm that the microbial and particulate levels are suitable for the intended use. The quality classification of the distribution line generally depends on the intended use (e.g., the classification of the air space within the equipment for which the CO₂ is used).
- b. The design of the distribution system should consider the pressure requirements of the intended use including any possible future expansion needs. Appropriate pressure regulation should be included in the design if equipment with different pressure requirements is connected to the distribution system.
- c. The design should include the necessary controls to provide a continuous supply of CO₂ gas (e.g., source tank switchover) at the appropriate flow rate (e.g., pressure design to achieve flow rate) and such features should be commissioned and/or qualified.

- d. Branched-distribution system designs serving different processes or different equipment within the process should consider how cross-contamination risk through the piping can be prevented.
- e. Pipes must be marked with CO₂ signs and should indicate the flow direction.
- f. An oxygen (O₂) sensor with visual and audible alarm should be installed in an appropriate location to notify personnel in the room, or those intending to enter, that a low-oxygen condition exists and that they should exit (or not enter) the space. Personnel entering the room should also be equipped with personal O₂ sensors. A buddy system and/or man-down alarms should also be considered.
- g. A risk-based approach should be used for the qualification.

3. Purified water generation and distribution system

- a. Any purified water generation and distribution system should be qualified and commissioned for the intended use. Routine sampling of both the generation and distribution system should be conducted to ensure the ongoing quality of the purified water.
- b. The use of purified water in cleaning activities should consider the impact of water quality on any surface-disinfectant preparation and on the ability to maintain room classification, or WFI should be considered as an alternative.
- c. The design and maintenance (e.g., temperature, flow rate, routine disinfection) of the distribution system should ensure no biofilm formation.
- d. Distribution systems serving different processes or different equipment within the process should ensure that the process, including the cleaning process, operates at a lower pressure than the purified distribution system to prevent a cross-contamination risk through the piping, or the valve design should prevent any possible mixing (e.g., double-block and bleed).
- e. Pipes must be marked with “Purified Water” signs and should indicate the flow direction.

4. Water for injection generation and distribution system

- a. Any WFI generation and distribution system should be qualified and validated to consistently produce water meeting the specification as it is intended to be used. Sampling locations close to the intended points of use should be defined according to the CCS. Routine sampling of both the generation and distribution system should be conducted to ensure the ongoing quality of the WFI.
- b. WFI should be produced by distillation or by a purification process that is equivalent to distillation if WFI is used for direct product-contact purposes.
- c. The design and maintenance (e.g., temperature, flow rate, routine disinfection) of the distribution system should ensure no biofilm formation.
- d. Distribution systems serving different processes or different equipment within the process should ensure that the process, including the cleaning process, operates at a lower pressure than the WFI distribution system to prevent cross-contamination risk through the piping, or the valve design should prevent any possible mixing (e.g., double block and bleed).
- e. Pipes must be marked with “WFI” signs and should indicate the flow direction.

5. Liquid nitrogen generation and distribution system

- a. Installation of a liquid nitrogen (LN₂) generation and distribution system should consider the asphyxiation risk. A room O₂-level monitoring system with visual and audible alarms should be available, and personnel should also be equipped with O₂ personal-monitoring instruments. Personnel in the room, or intending to enter the room, should be trained to exit (or not enter) the room when the alarm is active. A buddy system and/or man-down alarms

should also be considered. Means to detect and isolate any leak in the system should be available, including the availability of personal protective equipment (PPE) for routine use and emergency-rescue purposes.

- b. LN₂ is extremely cold and contact to bare skin causes severe frostbite. To mitigate this risk, signage should be posted throughout the space where LN₂ is in use that reminds operators of the appropriate PPE to don when handling materials immersed in LN₂ or when opening the equipment. In addition, PPE-donning stations should be prominently positioned within the floor plan. First-aid stations should also be available in the floor plan for accidental exposures.
- c. The materials that require storage in LN₂ (ultra-low temperature storage) are commonly the most critical materials in the process (e.g., cell or viral banks). Therefore, it is common in designs to consider the inclusion of back-up equipment (e.g., LN₂ freezers) in case of equipment failure. The back-up equipment should be qualified for use and subject to routine maintenance so it is ready for deployment should the primary equipment or primary equipment's controllers fail. This approach will require additional footprint and capital expenditure at the time of facility design, as well as appropriate piping and valving to supply LN₂ to both the primary and the back-up systems.
- d. Piping for distributing LN₂ should be adequately insulated both due to the cold-burn risk and to avoid condensation on external surfaces.
- e. Pipes should be marked with "LN₂" signs and should indicate the flow direction.
- f. Due to the risk of testing LN₂ and its stability, LN₂ quality through the distribution system is typically confirmed by the use of a Certificate of Compliance or Certificate of Analysis provided by LN₂ supplier.
- g. Design should include controls to provide continuous supply of LN₂ as required (e.g., source-tank switchover), and such features should be commissioned and/or qualified accordingly.

6. Nitrogen

- a. Any nitrogen (N₂) distribution line installed should be classified. The classification of the distribution line generally depends on the intended use (e.g., classified environment within the equipment where the N₂ is used, criticality of the material in contact with the N₂ gas). Examples of N₂ use include, but are not limited to, N₂ overlay in the product container head space and N₂ overlaying in the bulk-material containers (i.e., displacement of oxygen to reduce oxidation of materials and/or maintain the stability of raw materials).
- b. Installation of an N₂ generation and distribution system should consider the asphyxiation risk. An O₂-level monitoring system room with visual and audible alarms should be available, and personnel should also be equipped with O₂ personal-monitoring instruments. Personnel in the room, or intending to enter the room, should be trained to exit (or not enter) the room when the alarm is active. A buddy system and/or man-down alarms should also be considered. The means to detect and isolate any leak in the system should be available, including the availability of PPE for emergency-rescue purposes.
- c. Appropriately qualified source and terminal sterile filtration should be considered if N₂ is used for product-contact purposes.
- d. Distribution systems serving different processes or different equipment within the process should ensure that the design prevents the risk of cross-contamination.
- e. Pipes must be marked with "N₂" signs and should indicate the flow direction.

7. Oxygen

- a. Any O₂ distribution line installed should be classified. The classification of the distribution line generally depends on the intended use (e.g., O₂ sparging applications during cell expansion to maintain oxygen potential in the media/stock solutions).
- b. The design should include the control to provide a continuous supply of O₂ gas (e.g., source-tank switchover) at an appropriate flow rate (e.g., pressure design to achieve flow rate) and distribution uniformity.
- c. Appropriately qualified source and terminal sterile filtration should be considered if O₂ is used for product-contact purposes.
- d. Distribution systems serving different processes or different equipment within the process should ensure that the design prevents the risk of cross-contamination.
- e. Pipes must be marked with “O₂” signs and should indicate the flow direction.

8. Process air and process gas

- a. Any process air/process gas distribution (other than N₂ or O₂) line installed should be classified. Process air/process gas typically refers to air or gas intended for direct product-contact purposes (e.g., distributed into vessels containing product); therefore, the classification of the process air/process gas should be suitable for product contact. For such purposes, appropriately qualified terminal sterile filtration should be considered.
- b. For some gases (e.g., clean compressed air) it may be important to limit or reduce moisture or hydrocarbon content during and after generation, prior to the point of use. Consideration should be given to the cleanliness of the source (e.g., canisters, bulk supply, compressors) and alternatives should be sought that are of suitable cleanliness for the intended use (e.g., oil-free air compressors). Appropriate post-generation treatment (e.g., dryers, coalescing filters) should be specified for the process gas and the selected method of generation.
- c. Design should include the control to provide desirable flow rate and distribution uniformity during use.
- d. Distribution systems serving different processes or different equipment within the process should ensure that the design prevents the risk of cross-contamination.
- e. Pipes must be marked with “Process Air” signs and should indicate the flow direction.

9. Instrument air and compressed air

- a. Instrument air/compressed air typically refers to air used to support such processes as instrumentation use. Hence, the system is typically not considered as being in direct product or process contact. Nevertheless, the system should be qualified and commissioned.
- b. Pressure control should be instituted to ensure that the pressure at use is appropriate per instrument/equipment design.
- c. Pipes must be marked with “Instrument/Compressed Air” signs and should indicate the flow direction.

Rationale

Utilities used for the manufacturing of ATMPs should be qualified and validated, as in any other pharmaceutical product, in accordance with their intended use. If utilities such as liquid or gas (so-called “clean” utilities) come in contact with the pharmaceutical product, the quality should be demonstrated to meet CGMP requirements through both the system design and routine monitoring.

Topic 4. Process Equipment

Q 4.1. When choosing a biological safety cabinet for the manufacture of ATMPs, what factors are important?

Recommendation

A BSC is flexible in its use, although it typically does not have the same levels of integrated control and monitoring systems as an isolator. Instead, it relies on operators to manually decontaminate the Grade A space, maintain it through good aseptic practice, and appropriately monitor the environment. This increased reliance upon the operator means that operations should be carefully planned with an understanding of the barrier system limitations of the BSC and should incorporate appropriate risk mitigations to safeguard the aseptic operations. According to EU GMP Annex 1, the BSC is not classified as barrier technology.

The room environment where the BSC is located should support the maintenance of the Grade A environment within the operational zone for the barrier system in use. This dictates the need for a Grade B environment to surround the BSC.

Biosafety Cabinets

BSCs are widely used in the manufacturing of ATMPs as they are easy to install and maintain. Further, BSCs generally occupy a modest footprint that permits highly flexible operations within the enclosure due to the ease of access to the working area. BSCs allow access to the inner working zone via an open sash.

Note: BSC - Class III are the sole exception to the open-sash design. (Further descriptions of the different classes are provided in this section.)

Surfaces inside BSCs are typically cleaned and disinfected manually, which requires intensive training of operators to validate this procedure. Hence, BSCs require comprehensive QRM to ensure maintenance of aseptic conditions during all activities within the critical zone.

The open-sash operation of most BSC designs provides direct access by the personnel to the ATMP products being handled. Consequently, due to the exposure of personnel gowning to the products being worked within the BSC work zone, personnel gowning must support aseptic operations. The increased gowning requirements, as compared to those required for isolators, are more expensive and translate into a more complex and costlier EM program. Material transfer into the BSC poses a high risk, as transfers are done through the open sash, and materials are further decontaminated upon entry. The continuously open sash, with forearms and hands penetrating the opening to perform the manipulations, and the risk of airflow disturbance or air mixing during those manual operations contributes to the comparatively higher risk of BSCs.

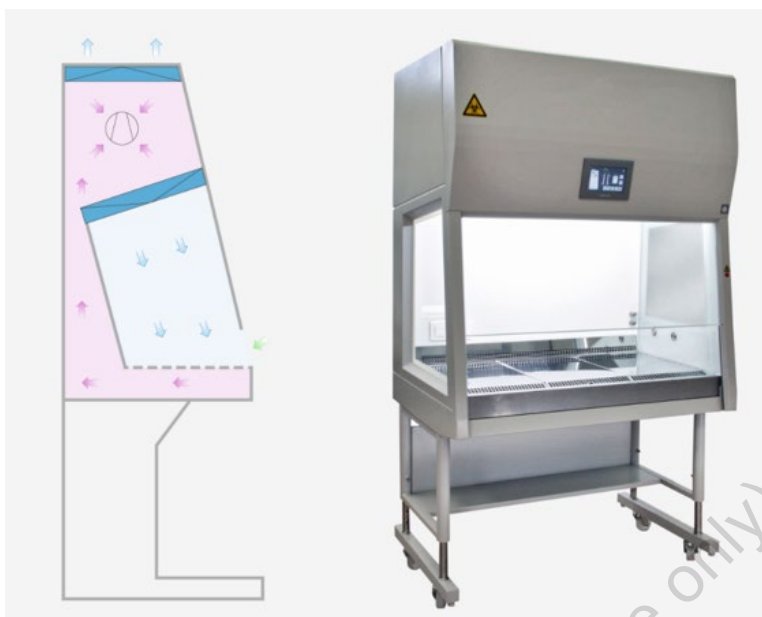


Figure 4.1-1 Biosafety Cabinet

The three main classes of BSCs are:

Class I BSC

Class I cabinets provide personnel and environmental protection by filtering air through a HEPA filter before it is exhausted. These cabinets do not protect the product from potential contamination and are suitable for general manipulations of low-to-moderate risk agents where product sterility is not a priority. Class I BSCs should never be used for ATMP-manufacturing activities.

Class II BSC

Class II cabinets are designed to protect the product, personnel, and environment, making them acceptable for handling ATMPs with a requirement for aseptic conditions. There are multiple subtypes of Class II BSCs, each with different operating conditions that dictates their suitability for the production of different classes of ATMP (see **Figure 4.1-2** below).

Class II Type A1 BSC

Often used in clinical and research laboratories for standard microbiological work with no chemical evaporation. Type A1 cabinets maintain a minimum inflow velocity of 75 feet per minute and recirculate 70% of the air while exhausting 30% outdoors after filtration. They are not suitable for handling volatile or toxic chemicals. Type A1 BSCs are usually only used in development laboratories where the ATMP will not be administered to a human.

Class II Type A2 BSC

Similar to Type A1 BSCs but with an increased inflow velocity of 100 feet per minute, enhancing containment. Type A2 cabinets can handle minute amounts of volatile chemicals if vented with an appropriate thimble (canopy) connection to the facility's HVAC system.

Class II Type B1 BSC

Designed for work that involves minute quantities of toxic chemicals and radionuclides, providing these operations occur toward the rear of the work area and are exhausted directly outside. Type B1 maintains a balance where 70% of the air is exhausted and 30% is recirculated, both through HEPA filters.

Class II Type B2 BSC

Also known as total exhaust cabinets, these are suitable for handling significant quantities of toxic chemicals and radio nuclides. They maintain a minimum inflow of 100 feet per minute and exhaust 100% of the air. All inflow and downflow air is removed via the ducts out of the building, allowing no recirculation within the cabinet.

Class II Type C1 BSC

A versatile cabinet that functions under both Type A and Type B conditions with a switch to change between modes depending on the chemicals used. This type of BSC provides flexibility in laboratories that handle both biological samples and chemicals, adjusting the exhaust accordingly.

Class III BSC

Offering the highest level of protection, Class III cabinets are gas-tight enclosures used with high-risk microbiological agents and hazardous chemicals. Materials are manipulated through attached gloves.

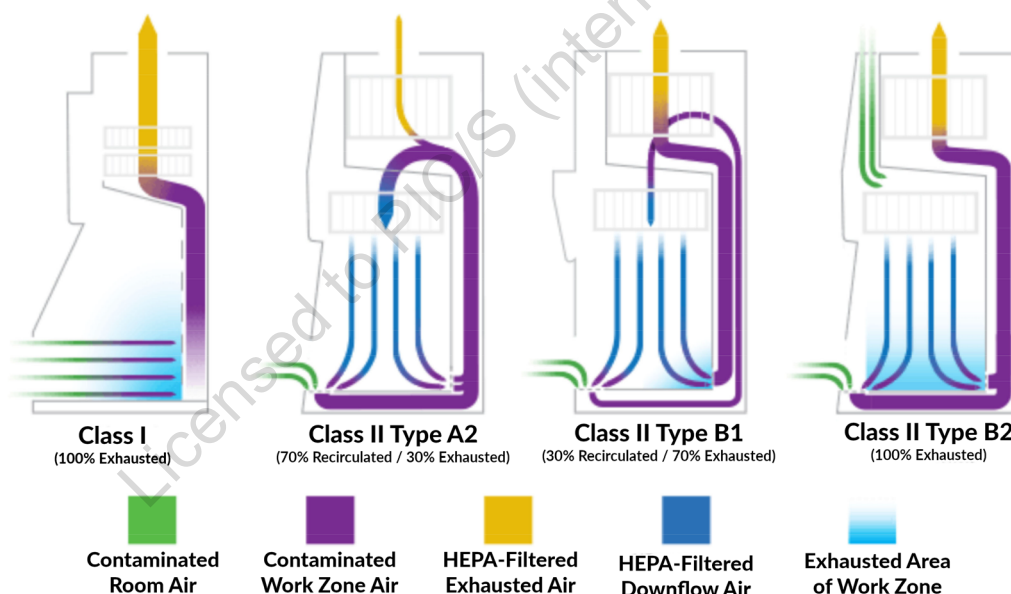


Figure 4.1-2 Airflow Pattern in a Biosafety Cabinet

Operating Considerations for the BSC

- Easy-to-perform manual processes through the sash opening
- Flexibility to adjust how the BSC is used based on the needs of the process
- Rapid changeover from process to process

Constraint Considerations for the BSC

- Material transfers can be difficult due to the need for manual sanitizing on the entry to the Grade A zone.
- An assistant is often needed to conduct the material transfer so the person who is manufacturing can continuously maintain their hands within the Grade A zone.
- Wrist rests should be purchased to avoid the operator covering the exhaust and effectively bringing the contaminated air into the Grade A environment.
- Long gloves are required as gowns cannot be sanitized upon entry to Grade A and should not enter this zone.

Licensed to PIC/S (internal use only)

Q 4.2. What factors are important when choosing an isolator for the manufacture of ATMPs?

Recommendation

An isolator typically has integrated monitoring of the Grade A environment using viable and non-viable (total particulate) monitoring in addition to air-velocity and pressure monitoring. This provides the ability to control and monitor the environment while the isolator is in use. This control and monitoring strategy should be planned and maintained. Isolators have the capability of undergoing an automatic decontamination cycle as they can be fully closed from the external environment. This ability during operations also provides effective separation from operators who can only interact with the interior through glove ports. Maintenance of Grade A conditions within an isolator is made simpler through this separation.

Isolators are considered closed barriers to the surrounding environment and may be placed in a Grade C or Grade D environment based on the CCS and the requirements of the market being served. In EU GMP Annex 1, closed isolators (examples **Figure 2.3-2**, **Figure 4.2-1**, and **Figure 4.2-2**) require a background of Grade D or higher and, for open isolators with small openings for a continuous or semicontinuous infeed/outfeed of materials, a Grade C background is required. The isolator system provides a Grade A work zone and should be equipped with an integrated, automated, and validated decontamination system (e.g., vaporized or nebulized H₂O₂).

Isolators

During operation, the operator's only access to an isolator is through the gloves mounted on the glass panels. The quality of the gloves, set up and design of the workspace, are important and offer flexibility. Working with gloves through the barrier might be more challenging and could require enhanced training to perform sensitive work, which could result in longer operation times.

According to the EU GMP Annex 1, isolators are classified as a barrier technology. Open or closed isolators may be attached to adjacent chambers, permitting the closed transfer of materials in and out, as shown in **Figure 4.2-1** with a MAL to the left of the main isolator chamber. The MAL system provides a fast method for decontamination of materials in and out via a closed and interlocked chamber which maintains the Grade A classification within the isolator. The VPHP used in the MAL decontamination cycle might pose a risk to the transferred material and should be carefully assessed. Additional process technologies, such as an incubator, centrifuge, or robotic system, can be attached and integrated within an isolator system to provide a closed and continuous working environment.



Figure 4.2-1 Closed Isolator with Material Air Lock, Centrifuge, and Incubator

The permissible maximum leak rate of the isolator should account for the criticality of the manufactured product. As a recommendation, the leak rate should be a maximum of 5% volume rate per hour (the typical capability of today's systems often falls within 3–5%).

A routine monitoring program should include frequent leak-testing of the isolator and integrity-testing of the glove/sleeve system. Supporting data from leak-testing should align with segregation policies and compliance requirements.



Figure 4.2-2 An Example of a Closed Isolator System for the Manufacturing of ATMPs

Computational fluid dynamics (CFD) studies are powerful tools to predict the spreading of particles and aerosols. Particles and aerosols should be transferred to air-return duct filters in the fastest and shortest pathway possible. The design process should make use of appropriate engineering controls, such as CFD (**Figure 4.2-3**), to demonstrate the control of the airflow towards the safety filters within the air-return ducts (**Figure 4.2-4**). The criticality of the manufactured ATMP should always drive the necessary design attributes for both tightness (i.e., leaks) and air flow controls. For potent products, additional HEPA filters should be fitted before the air-return ducts to avoid spreading particles and aerosols outside the isolator chamber.

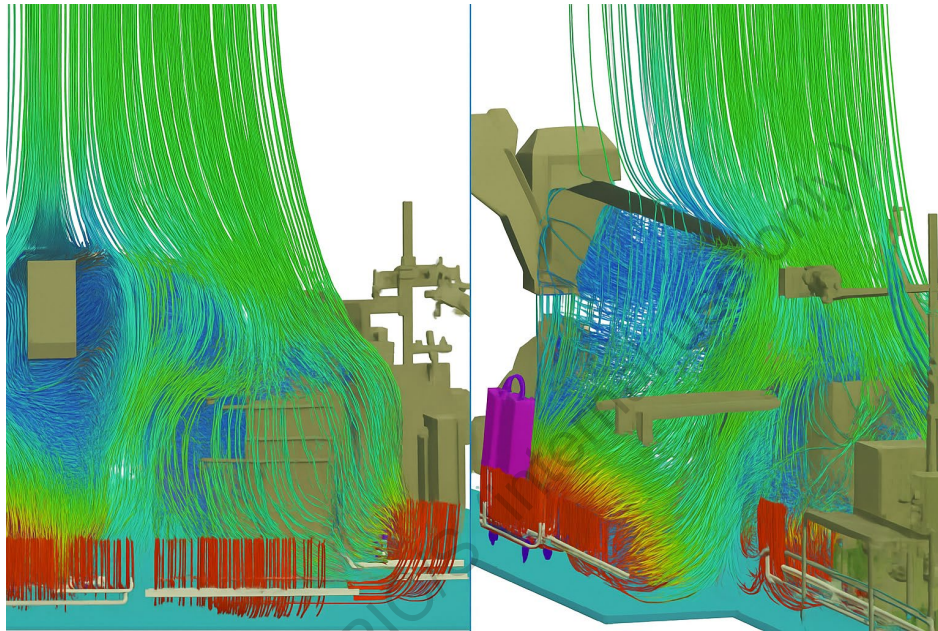


Figure 4.2-3 Computational Fluid Dynamics Studies



Figure 4.2-4 Safety Filter Before the Air-return Ducts

Operating Considerations of Isolators

- Provides controlled closed environment to protect the product and the operator.
- Surface decontamination with vaporized hydrogen peroxide (VPHP) or other decontamination agent systems is automated and validated.
- Controlled material transfer in and out of the isolator system through RTPs or MALs with automated, validated decontamination cycles (see **Figure 4.2-1**).
- Isolation of the pharmaceutical product within a small, closed environment that avoids spreading highly potent pharmaceutical products to surrounding areas.
- Higher BSL requirements can be accommodated due to the closed environment, automated decontamination, and the possibility of inactivation.
- More sustainable due to lower surrounding cleanroom environment (Grade D), which means less room EM, less gowning, and less related training.

Constraints of Isolators

- Isolators require careful design to ensure that all required aspects of the process can be executed through the designated glove ports and to verify that all required material transfers can occur through the designated RTPs or MALs.
- Isolators may, in some circumstances, be less flexible in terms of significant process or equipment changes.
- Isolators require routine leak-testing of the chamber(s) and integrity-testing of the gloves, sleeves, and glove ports to ensure that the Grade A environment is maintained.

Q 4.3. What should be considered when comparing biosafety cabinet and isolator design?

Recommendation

The design of an ATMP facility needs to be focused first considering its containment. The design of the containment and its background environmental classification are key considerations that will drive all other design elements (see **Table 4.3-1**). The choice of containment measures will be based on the existing facility design constraints, size and layout of the facility, new build budgets, and other company activities (see **Table 4.3-2**).

Regulators are increasingly pushing industry for containment measures similar to isolators, even if processing is not performed within an isolator.

Table 4.3-1 Background Grades and Assurances Considerations

Primary Containment – Grade A	Background Requirements	Considerations Summary (not exhaustive)
Biological Safety Cabinet (BSC) Class II	Grade B	- Only partial barrier with processing occurring through the open sash, with gownned forearm and gloved hand exposure to the inside of the BSC. - Manual decontamination. - Sanitization of items (and hands) upon entering the Grade A environment each time. - Wrist rests recommended to fix in place the operator's arms and assist in reproducible airflow patterns (and to avoid arms resting on the air exhaust). - Considered high-risk by regulators due to the close proximity of the operator to the process and the only barrier being the operator's clothing within the Grade A.
Isolator – Closed	Grade D*	- Complete barrier between operator, and the process is fixed and sealed. - Decontamination of items occurs during a validated decontamination/sanitization cycle. - Gloves can be reliably integrity-tested using automated methods. - Glove ports are set in place, so airflow movement is well-controlled. - Considered as the higher sterility assurance provider by regulators due to automation of decontamination/sanitization and increased physical barriers.
*Note: Grade D is the minimum grade expected. Grades can be higher depending on the nature of the ATMP and the need for low-particulate transfer (e.g., such as during isolator loading operations, if any) as defined by the CCS.		

Operational Comparison of Biosafety Cabinets and Isolators

BSCs and isolators are typically equipped with a filter of at least a HEPA classification. The type of HEPA filter installed is expected, at a minimum, to have particulate removal of 99.99% when tested, in accordance with internationally acceptable standards. The filtration capability of HEPA filters in BSCs and isolators should provide a Grade A operating work zone.

Since a BSC is operated via an open sash compared with the closed operating mode of an isolator, there is a greater need to manage particulates and contaminants through good aseptic practice. Products in the work zone of the BSC may be exposed to particulates from gowning worn by personnel since the personnel are required to manipulate the products directly via gloved hands.

In contrast, an isolator provides a closed environment where personnel indirectly access the work zone through integrated gloves. The use of gloves attached to the isolator may make product-handling inconvenient (i.e., due to the loss of some physical dexterity) in an isolator compared to a BSC. However, indirect access not only protects the product and process but also protects the operator as there is no risk of transfer of contamination to the operator's gowning.

Production within isolators also reduces the gowning requirement for operators since they may be gownned in accordance with the requirements of the background environment. This means operators working with isolators may be gownned for Grade D (or C) operations, while operators working with BSC must gown for Grade A operations. The lower gownning requirements can save some expense in training, gownning, time, and EM, making the isolator more sustainable over the long term.

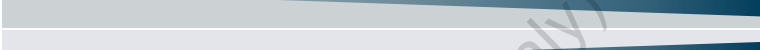
The surface cleaning and decontamination/disinfection in a BSC is done manually using sterile wipes. In an isolator, the cleaning is performed manually while the decontamination is performed with an integrated, automated, and validated technology such as vaporized or nebulized H₂O₂ or another decontamination agent.

Material transfers in a BSC are performed with the open sash, as one operator transfers items to the operator working in the BSC. During this process, it is important to avoid contamination of the surrounding area with the material transferred through the open sash by careful decontamination of external surfaces and/or removal of packaging. Transfer technologies attached to the BSC, such as RTPs, can provide lower-risk transfers. Isolators use transfer systems like MALs and/or RTPs to provide a safe and rapid transfer after a vaporized or sprayed decontamination agent cycle. The exposure of the decontamination agent to the materials being transferred should be investigated for chemical and penetration capability.

According to EU GMP Annex 1, the BSC is not classified as a barrier technology, whereas isolators are.

Comparison Between Biosafety Cabinet and Isolator for Facility Design

Table 4.3-2 Comparison Between Biosafety Cabinet and Isolator for Facility Design

Facility Design	Class II BSC	Isolator
Contamination Control		
Operator Protection		
Sustainability		
GMP Annex 1 Compliance		
Cleanroom Footprint		
Integration into an Existing Cleanroom		
Equipment Lead Time		
Key: Lowest AdvantageHighest Advantage		

Capital Expenditure and Operating Cost Considerations

The capital expenditure (CapEx) and operational expenditure (OpEx) costs for BSCs and isolators need to be evaluated considering the individual circumstances (see **Table 4.3-3**), such as:

- Manufacturing time (short or long)
- Complexity of the process
- Availability of trained and experienced personnel for the selected technology
- Degree of automation (e.g., robotics)

An isolator system is typically more complex than BSC, and the complexity contributes to the higher cost of purchasing an isolator system (see **Table 4.3-4**). Similarly, this complexity translates to higher operational and maintenance costs for the equipment (e.g., parts replacement, complex filter design, enhanced training for the skillset required for operation), but lower investment costs for the secondary containment, as the isolator system is placed in a minimum of a Grade D environment.

Table 4.3-3 Capital Expenditure and Operating Cost Considerations

Primary Containment – Grade A	Background Requirements	Considerations Summary
Biological Safety Cabinet (BSC) Class II	Grade B	High OpEx and Low CapEx
Isolator – Closed	Grade D*	High CapEx and Low OpEx
*Note: This is the minimum grade expected. Grades can be higher depending on the nature of the ATMP and the need for low particulate transfer as defined by the CCS.		

Cost Comparison Between the Use of Biosafety Cabinet and Isolator

Table 4.3-4 Cost Comparison Between the Use of Biosafety Cabinet and Isolator

Consideration	Cost	Class II BSC	Isolator
Equipment	Initial Investment (capital)		
	Low-volume Products (operational and maintenance costs)		
	Medium/High-volume Products (operational and maintenance costs)		
Room	Initial Investment (capital)		
	Operational and Maintenance Costs		
Gowning	Gowning Cost		
	Environmental Monitoring Cost		
Key: Lower Cost Higher Cost			

Facility Layout Considerations

As the BSC requires a Grade B background and the isolator requires a Grade C/D background (CCS-dependent), the room layout of the facility will be different to fulfill CGMP requirements. Additional considerations, such as the BSL necessary for the product, may also have an impact on the facility layout design. It may include a containment area surrounding the BSC or isolator room provided by a relatively negative pressure directly adjacent to a higher positive-pressure area, causing a “sink” or “bubble,” in order to prevent transfer into other parts of the facility. An example of a layout for a production suite with an installed BSC compared to a production suite with an isolator is shown in **Figure 4.3-5**.

Small Scale, ATMP Manufacturing Air-flow Diagram

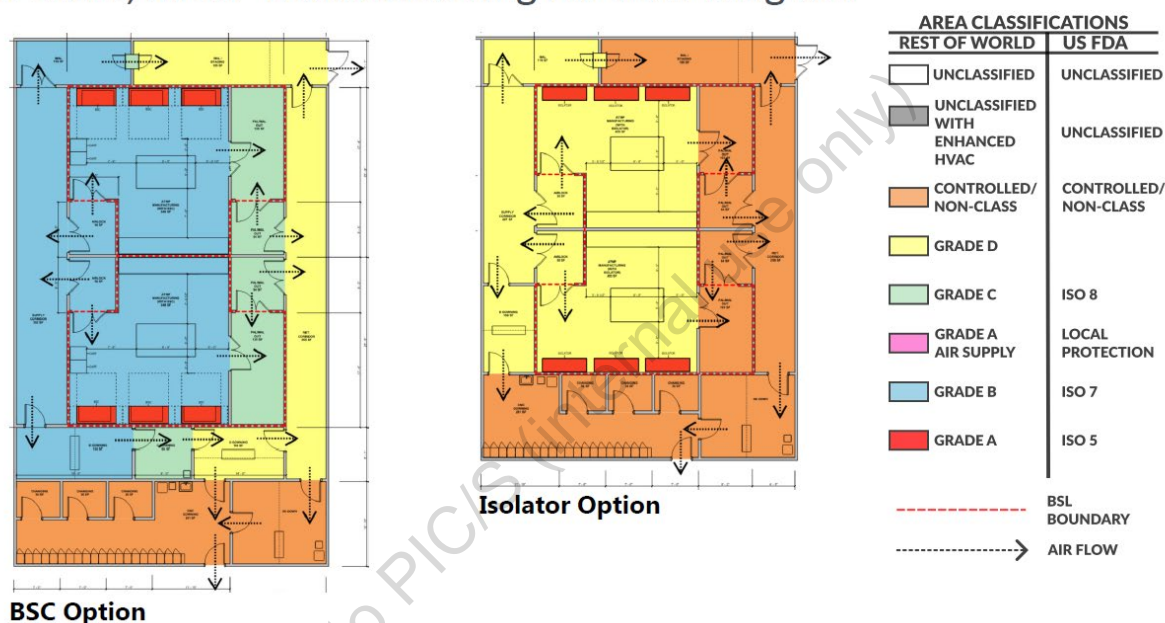


Figure 4.3-5 Isolator and Biosafety Cabinet Environment Options (positive-pressure cascade and bubble in the air locks)

Rationale

Since most ATMPs should be manufactured under aseptic conditions to protect them from particulate or microbiological contamination, barrier systems are recommended. Isolators can provide improved sustainability benefits and compliance with EU GMP Annex 1. However, BSCs are often used as they offer flexibility due to their open design, especially during the early stages of process development. For working with viral vectors and/or multiple products, an isolator provides a barrier that isolates the highly potent product from both the operator and the environment. Furthermore, an isolator provides an automated and validated solution to inactivate any potent product remaining after the end of the batch or campaign (e.g., H₂O₂ decontamination). Costs may or may not be a discriminating factor as the initial CapEx savings of BSC are often offset by the higher lifecycle costs of the OpEx.

Topic 5. Supplementary Technical Information

Q 5.1. What is important to consider in a mock-up for biosafety cabinets and isolators?

Recommendation

A mock-up for the BSC or isolator at the vendor's site should be built to ensure that all reasonably practicable steps have been taken to ensure good aseptic practice when performing the necessary motions for the process, ergonomics for the operators, and the ability to perform the required maintenance. With an isolator system, the vendor normally performs a mock-up with wood, cardboard, or similar rigid materials because, after the isolator system is installed, the glove position can only be changed with a significant effort and cost. With BSCs, the designs are normally standard and vary only with the size of the BSC.

Nevertheless, it is important to ensure that the process can be performed with good aseptic technique within the size constraints of the selected BSC, and so it might be most appropriate to perform with a mock-up (or an actual unit) of the intended size.

The mock-up exercise should consider the following:

- Working position, including whether the operator will be seated or standing.
- Positions of all equipment within the Grade A area.
- Positions of all components and materials including hangers, trays, or stands required to support materials.
- Position of process waste.
- Location for all product-contact tools before and during the process, with special consideration for ensuring that they are continuously bathed in first air and are not touching any potentially contaminated surface at any time.
- Process sequence/flow and all required manipulations and interventions.
- Location of critical surfaces relative to first air and required operator movements.
- Locations of continuous viable and nonviable sampling in the Grade A area to accurately capture any activity within the working position and to ensure that EM materials, where required, can be introduced and placed within the working zone without breaking first air over the process or process equipment.
- Operator hand/arm/glove movements throughout when considering the locations of the working position to ensure the operator's hand/arm/glove does not move over critical surfaces.
- All required manipulations and variations in setup, including loading and unloading of any materials from equipment, air locks, or RTP, should be tested by both right- and left-handed individuals, and the testing should include personnel of varying heights and arm lengths to ensure that the configuration will be suitable for a wide range of operators performing the process.
- Frequency and duration of operations.
- Weight/force for each action.
- Setup and performance of cleaning and disinfection actions.

For the mock-up, it is highly recommended to involve the operator(s) responsible for handling the equipment in routine production. The mock-up process should be conducted by a minimum of two trained operators so the second person can observe and record actions or violations that may not be apparent to the person conducting the process. The results from the mock-up should be documented.

The mock-up may be performed using simplified representations of tools, materials, and equipment manufactured from basic materials to aid in rapid, low-cost iteration. This use of a low-cost model is especially worthwhile during the initial process or initial isolator/BSC design and layout when there may be multiple iterations of the design. However, for the final mock-up testing, it is important to ensure that the selected representations are adequate to challenge the weight/force that may be required for the actual manipulations.

Besides rigid mock-up studies, computerized mock-ups and virtual-reality mock-ups can also support the physical mock-up (see **Figure 5.1-1**, **Figure 5.1-2**, and **Figure 5.1-3**). Physical, computerized, or virtual mock-ups are often retained by the operating company in order to be used to test new process movements or interventions before incorporating them into the process or to provide preliminary training to new operators. As a consequence, cost spent on making the model realistic can be an investment in the process maintenance over time.



Figure 5.1-1 Computerized Mock-Up

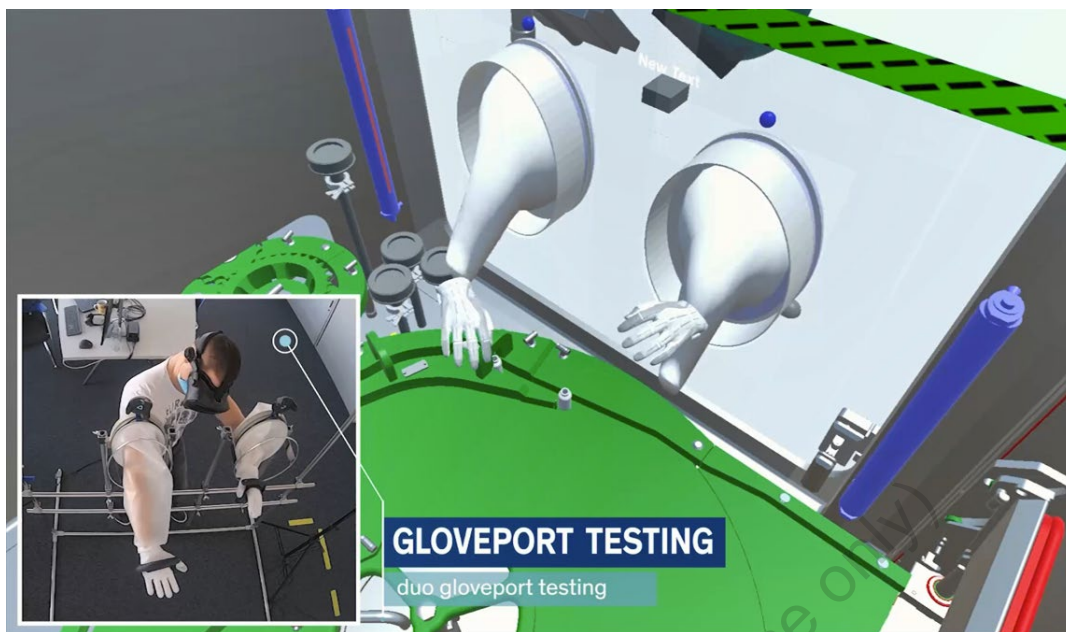


Figure 5.1-2 Virtual-Reality Mock-Up



Figure 5.1-3 Physical Mock-Up

Rationale

The manufacture of ATMPs commonly involves many complex sequences of operations that may be performed manually by operators. This combination means that, without careful planning and design of the aseptic operation, there is a serious risk of failure. Ensuring that the equipment design is compatible with and optimal for the intended process is highly important to the ultimate success of the equipment–process pairing. Good aseptic practice may conflict with the quickest or simplest manufacturing process and, therefore, without careful planning and design of the process operations via a mock-up, the default process may compromise sterility. The physical and virtual-reality mock-ups also provide an opportunity to define, refine, practice, and train on the required aseptic processing procedure.

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Q 5.2. Are there any differences in the frequency of chamber leak-integrity testing procedures between an isolator and a biosafety cabinet?

Recommendation

For isolators, the leak test should be performed frequently with the interval established, based on the frequency of intended operations in an appropriate risk assessment **(1, 7-10)**.

The leak-testing frequency for isolators should not be limited to periodic evaluation, but rather should include such situations as:

- After changing gloves and sleeves (documented in the glove change management protocol)
- Following any maintenance or repair that has breached the sealed structure
- Supporting failure investigations, especially those involving viral and/or bioburden contaminations

PDA *Points to Consider for the Aseptic Processing of Sterile Pharmaceutical Products in Isolators* (2020) recommends that the leak-testing (integrity) be performed at defined periods, at a minimum, at the beginning and end of each batch or campaign and following any intervention that would affect the integrity of the unit **(10)**.

The testing procedures performed for leak integrity in isolators are described in *Annex E.3 Quantitative leak testing of ISO EN 14644-7*. Table E.1 in ISO 14644-7 indicates the oxygen method (only for negative pressure), the pressure-change method, or the Parjo method (very rarely used; mainly in atomic industries) as being appropriate for very tight ISO containment of Class I devices **(11)**. In addition to a tightness classification, *ISO 10648-2* describes three different methods for leak-testing negative-pressure, rigid-wall separative devices as the oxygen method, the pressure-changes method, and the constant-pressure method **(12)**. The oxygen, pressure-change, and constant-pressure methods are listed as appropriate test methods for isolators with ISO containment Class III. Apart from other methods, PDA *Technical Report No. 34: Design and Validation of Isolator Systems for the Manufacturing and Testing of Health Care Products* also suggests leak-testing using a tracer gas, in which a tracer gas is released inside the isolator, and testing for the presence of the gas is performed outside the isolator **(9)**.

As BSCs are an open system, a leak test cannot be performed before each batch as is done with an isolator system. A containment test method, as shown in **Figure 5.3-1** and **Figure 5.3-2**, could be used instead of the leak test. This testing is typically performed upon first commissioning and validation for use and may be repeated periodically (e.g., annually or biannually) to check on the continued performance of the system, or when required for significant investigations into the system's operation.

Q 5.3. What are the similarities and differences between a biosafety cabinet and an isolator in the objective, procedure, documentation, and expectations of an airflow visualization (smoke study)?

Recommendation

The airflow visualization, or smoke study, is a method used to aid in the evaluation of the CGMP-required airflow generated by a BSC, isolator system, or containment enclosure. Common criteria for these studies commonly include such elements as unidirectional airflow in Grade A areas and airflow around the apertures between Grade A and lower-grade areas. Differences in the application of airflow visualization arise as a result of the intended purpose of the equipment. For example, some systems are intended for the protection of the operator from the product while other systems are meant for the protection of the product from external contamination.

For isolator systems, the airflow is designed to ensure that the first air from the HEPA filter flows over the sterile product and sterile equipment used in its manufacture. This is achieved by unidirectional airflow over the product and associated critical surfaces. In isolator systems, the objective of the airflow visualization is to verify that the airflows create a “sweeping action over and away from the product under dynamic conditions” as defined in the *FDA Guidance for Industry: Sterile Drug Products Produced by Aseptic Processing—Current Good Manufacturing Practice*, EU GMP Annex 1, PIC/S Annex 1: *Manufacture of Sterile Medicinal Products*, and WHO TRS 1044 - Annex 2: *WHO Good Manufacturing Practices for Sterile Pharmaceutical Products (13-15)*.

The airflow visualization procedure (smoke study) demonstrates that the airflow meets the design objectives. The study is focused on aiding in the visualization of the air as it moves down from the filter face over and away from the critical product and material surfaces. It ensures there are no airflows towards the product from sources that may generate particulates/contaminants (e.g., manual operations, moving machine parts). The airflow visualization studies include visualization of flow at the following points:

- Downflow across the chamber/work zone at the working position.
- Airflow with sweeping action over and away from the product.
- Airflow around large objects that may generate turbulence.
- Airflow, for example, during manual operation, interventions, and transfers that demonstrate the absence of turbulence on critical surfaces and that first air is not disturbed.
- Airflow during the processing of open units demonstrates that no air turbulence enters open units.
- Airflow at apertures between zones of differing pressure that demonstrates flow from the critical area outward without reflux or entrainment.

Airflow visualization should be performed at rest and in operation, including manufacturing setup in Grade A zones and during the performance of interventions. The studies should demonstrate that air does not ingress from the lower-grade cleanroom/zone to a higher-grade cleanroom/zone.

For BSCs, the airflow visualization mandated by the regulatory design standards specifications still applies. For example, *NSF/ANSI 49 – 2019 Biosafety Cabinetry: Design, Construction, Performance, and Field Certification* requires the following airflow visualization be successfully performed at least annually (16):

- Airflow work zone along the centerline of the work surface is downward.
- Airflow along the entire perimeter of the sash at the outside of the cabinet is inward into the grill before the work zone and does not intrude into the work zone.
- Airflow behind the sash, on top of the access opening, does not escape from the BSC.

- Airflow behind the window, at the sides and top of work area, is not leaking out of the BSC.

At all times, for visualization in either BSC or isolators, the actual setup of the process equipment, materials, and product should match the intended operations. The airflow visualization testing should comply with the requirements defined previously in this section and, in addition, should ensure that there are no obstructions to airflow grills.

For BSCs as containment enclosures, the primary purpose is to ensure that any hazardous product material liberated in the airflow is transported away from the operator to prevent the risk of harm. This airflow visualization is used mainly for semi-open Class I or Class II BSCs, according to *BS EN 12469 Biotechnology - Performance Criteria for Microbiological Safety Cabinets*, and does not quantify the potential contaminants. The procedure, therefore, demonstrates that the airflow meets the objective and is focused on aiding visualization to ensure there are no airflows towards the cleanroom, surrounding environment, and operator (17). This includes visualization of flow at the following points:

- Airflow around large objects that may generate turbulence.
- Airflow around continuous viable and nonviable sampling equipment.
- Airflow around hand movement and equipment while performing required operations.
- Airflow during interventions.
- Airflow remaining undisturbed by external airflow from the surroundings.

The documentation of the airflow visualization should be similar regardless of the objective of the study. Visual recordings of the process should be made from as many angles as are necessary to capture the flow (first air) at each critical point, including in dynamic conditions (i.e., when the system is in use). The FDA and European Medicines Agency recommend that the documentation provide a detailed summary of what equipment and supplies are present, the number of personnel present, and a list of all activities simulated.

In addition to airflow visualization studies, there are other methods that are commonly employed for BSC that are nonvisible but use a variety of detectors in order to classify the performance of the system in quantitative terms. These tests include, but are not limited to, a segregation test, KI-Discus (potassium iodide) test, or other containment tests.

Unexpected airflows between clean zones can also be detected using the segregation test, according to Chap. 4.2.11 of *ISO 14644-3:2019 Cleanrooms and Associated Controlled Environments, Part 3: Test Methods*. The lesser classified area is challenged with a defined concentration of aerosol particles, and the particle concentration in the protected area is then measured. The difference between the two concentrations results in the determination of a protection factor (18).

For hazardous applications in semi-open BSCs, the KI-Discus test method, according to EN 12469, is routinely used as a containment test. This relatively static test uses small salt particles from potassium iodide with a defined momentum to show the concentration difference between the interior of the BSC and exterior zone in front of the BSC, expressed as a “protection factor” (see **Figure 5.3-1**).



Figure 5.3-1 Containment Test Method on a Biosafety Cabinet (potassium iodide test)

For hazardous applications, fume cupboards can be evaluated using a quantitative containment test wherein an alcohol dispersion with a defined momentum and a defined concentration is measured instead of potassium iodide. On the outside of the cabinet, the alcohol concentration is measured around the dynamic horizontal movement of a dummy equipped with two metal arms in the workspace (see **Figure 5.3-2**). After a concentration analysis is performed, a containment performance class may be defined.



Figure 5.3-2 Containment Test Method on a Fume Cupboard (alcohol dispersion test)

Rationale

Airflow visualization, in its many forms, is an effective way to demonstrate that the process and product are protected by the designed unidirectional airflow from adverse influences such as particle-generating activities or the external environment. Since air in its natural state cannot be seen, the introduction of tools that enable its visualization (and, if needed, the quantitation of a tracer compound) provides a powerful mechanism to confirm the initial conclusions from the CFD models used during design. When performed under all of the conditions that represent routine production and interventions, it can be demonstrated that the intended process does not contribute additional risk to the environment through the disruption of airflow.

Q 5.4. What methods are used to maintain facility cascade/pressure stability?

Recommendation

Pressure cascades can be measured using two methods. The first method is to directly measure the pressure difference between two adjacent areas where common areas exist. For ease of validation, it is recommended that individual tubing be used between each high- and low-pressure sample-point location, although it is also possible (albeit more complex) that the monitoring points be connected together via a manifold. The advantage of this first method is a direct response to a pressure cascade change of state. The second method is to measure the absolute pressure of each space and calculate the differential. The advantage of this method is typically the use of fewer sensors and a wider view of the comparison of pressures between multiple spaces across the facility. However, this advantage is offset by a wider error range for measurement, as two sensors are required for the calculation of each differential. In either case, a loss of local pressure would be identified (see **Figure 5.4-1**).

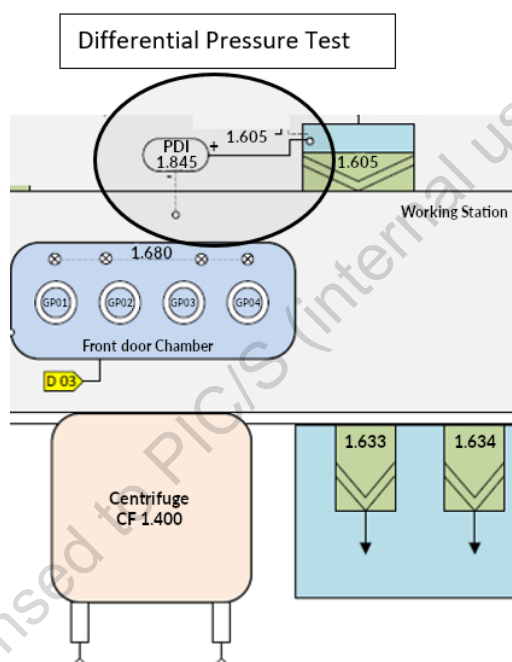


Figure 5.4-1 Shows the Pressure Difference Measurement Between the Isolator Chamber and Above the Supply HEPA Filter

A suitable measurement interval for pressure readings should provide a balance between small variances caused by internal fluctuations within a controlled space and the loss of control parameters, while enabling the averaging of results accurately over longer periods. The process and use of the cleanrooms (e.g., entries, exits, frequency of use, movements) should be used to determine the measurement and recording frequency as this may be different throughout each zone within the facility.

If a pressure loss has been observed, the air-handling unit should react immediately to balance the pressure loss. Pressure loss should be recorded, and an alarm should signal. The reason for the pressure loss must be investigated. Alarm signals should be timed to avoid signaling for routine operational events such as entries and exits.

Manual operation in BSCs and isolators, or the opening of a door at a PAL or MAL, are the most critical factors of an influence on the air stability in a system. Fast movements should be avoided. A higher degree of automation could reduce or even eliminate manual operations and the resultant air influence. Delays in alarms should be minimal (i.e., < 3 min) and should be justified as part of the CCS.

For cleanrooms, air locks with door interlocks on all doors and pressure stabilization are recommended to avoid pressure fluctuations between the air lock and the processing room. Furthermore, disruption of the differential pressure between the air locks and the adjacent cleanrooms will disrupt the function of the air locks as a pressure sink (negative relative to surrounding spaces), air bubble (positive relative to surrounding spaces), or pressure cascade (positive on one side and negative on the other side). (See **Figure 4.3-1.**)

Rationale

Stable pressure zones and maintaining the differential pressure between adjacent spaces are important for the safe manufacture of ATMPs and are essential to keeping aseptic conditions and preventing contamination and cross-contamination in adjacent areas. A pressure-zone concept, with a risk evaluation based on the assigned function of every room, as well as the personnel and material flows, should be part of the overall layout of the rooms within the facility.

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Recommended Reading

The following materials are suggested as additional reading with relevant information and regulatory guidance from health authorities and references to applicable publications or PDA technical documents.

American Society of Mechanical Engineers

American Society of Mechanical Engineers. *ASME BPE-2022 (Revision of ASME BPE-2019): Bioprocessing Equipment*; ASME: New York, NY, 2022.

ASTM International

ASTM International. *ASTM E2500-20 Standard Guide for Specification, Design, and Verification of Pharmaceutical and Biopharmaceutical Manufacturing Systems and Equipment*; ASTM: West Conshohocken, PA, 2020.

European Commission

European Commission. *Directive 2001/83/EC of the European Parliament and of the Council of 6 November 2001 on the Community Code Relating to Medicinal Products for Human Use*; OJ. Publications Office of the European Union: Brussels, 2001.

European Commission. *Guidelines on Good Manufacturing Practice Specific to Advanced Therapy Medicinal Products, EudraLex – Volume 4 – Good Manufacturing Practice for Medicinal Products for Human and Veterinary Use*; European Commission: Brussels, 2017.

European Medicines Agency

European Medicines Agency. *Guideline on the Quality of Water for Pharmaceutical Use*; EMA: London, 2020.

International Council for Harmonisation

International Council for Harmonisation. *Quality Guideline Q9(R1): Quality Risk Management*; ICH: Geneva, 2023.

International Organization for Standardization

International Organization for Standardization. *ISO/TS 23565:2021 Biotechnology — Bioprocessing — General requirements and considerations for equipment systems used in the manufacturing of cells for therapeutic use*; 2021.

International Organization for Standardization. *ISO 10648-1:1997 Containment Enclosures — Part 1: Design Principles*; 1997.

International Society for Pharmaceutical Engineering

International Society for Pharmaceutical Engineering. *Baseline Guide Vol. 3: Sterile Product Manufacturing Facilities, 3rd Edition*. ISPE: Bethesda, MD, 2018.

International Society for Pharmaceutical Engineering. *ISPE Baseline® Guide Vol. 5: Commissioning and Qualification, 2nd Edition*; ISPE: Bethesda, MD, 2019.

International Society for Pharmaceutical Engineering. *ISPE Good Practice Guide: Sampling Pharma Water, Steam, & Process Gases*. ISPE: Bethesda, MD, 2016.

International Society for Pharmaceutical Engineering. *ISPE Good Practice Guide: Good Engineering Practice, 2nd Edition*. ISPE: Bethesda, MD, 2016.

International Society for Pharmaceutical Engineering. *ISPE Good Practice Guide: Process Gases, 2nd Edition*. ISPE: Bethesda, MD, 2023.

International Society for Pharmaceutical Engineering. *ISPE Good Practice Guide: Approaches to Commissioning and Qualification of Pharmaceutical Water and Steam Systems, 3rd Edition*. ISPE: Bethesda, MD, 2016.

International Society for Pharmaceutical Engineering. *ISPE Good Practice Guide: Critical Utilities GMP Compliance*. ISPE: Bethesda, MD, 2020.

International Society for Pharmaceutical Engineering. *ISPE Guide: Advanced Therapy Medicinal Products – Allogeneic Cell Therapy*. ISPE: Bethesda, MD, 2024.

International Society for Pharmaceutical Engineering. *ISPE Guide: Advanced Therapy Medicinal Products – Autologous Cell Therapy*. ISPE: Bethesda, MD, 2021.

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